Trajectories of weak-field orbital evolution via gravitational radiation reaction in Schwarzschild - de Sitter spacetime

In this notebook we show the plots of orbital trajectories for compact objects around Schwarzschild black holes influenced by an SdS parameter λ (y in this notebook). Results of this notebook has been used in Villanueva, John Adrian N., and Ian Vega. “Extreme-mass ratio inspirals in Schwarzschild-de Sitter spacetime I: Weak-field orbits.” arXiv preprint arXiv:2602.17154 (2026). https://arxiv.org/abs/2602.17154

We solve for the system of osculating orbital parameters,

$$\frac{d\psi}{dt} = n(p)(1+e\cos\psi)^2 \tag{3.33}$$

$$\frac{d\delta}{dt} = \frac{3n(p)}{p} \tag{3.34}$$

$$\begin{aligned}\frac{dp}{dt} &= -\frac{64}{5}p^{-3}(1-e^2)^{3/2}\left(1+\frac{7}{8}e^2\right) \\ &\quad -\lambda\left[16(1-e^2)-\frac{4\left(26e^4-283e^2-168\right)}{15\left(1-e^2\right)^{3/2}}\right. \\ &\quad \left.+\mathcal{O}(\lambda p)\right]\end{aligned} \tag{3.35}$$

$$\begin{aligned}\frac{de}{dt} &= -\frac{304}{15}p^{-4}e(1-e^2)^{3/2}\left(1+\frac{121}{304}e^2\right) \\ &\quad -\lambda\left[\frac{32}{3}p^{-1}e^{-1}\left(1-e^2\right)^{3/2}\left(1-(1-e^2)^{1/2}\right)\right. \\ &\quad -p^{-1}e^{-1}\frac{2\left(73e^6-784e^4-965e^2-24\right)}{15\left(1-e^2\right)^{3/2}} \\ &\quad \left.+\mathcal{O}(\lambda p^{-2})\right].\end{aligned} \tag{3.36}$$

```
ClearAll["Global"]
```

In[ ]:=
```
SetDirectory[NotebookDirectory[]];
```

In[7]:=
```
plotsFolder = FileNameJoin[{ParentDirectory[NotebookDirectory[]], "plots"}];
(*automatically saves to the plots folder in the repo*)
```

In[8]:=
```
Get["MaTeX`"];(*MaTeX for plot generation*)
SetOptions[MaTeX, FontSize → 19,
  "Preamble" → {"\\usepackage{newtxtext,newtxmath}"}, Magnification → 1.2];
style = Directive[FontFamily → "Times", FontSize → 23, Black];
cm = 72 / 2.54; (* 72 points per inch and 1 inch = 2.54 cm *)
```

Orbital equations:

In[1]:=

$$\text{fdot}[p_, e_, f_, y_] := \left(\sqrt{\frac{1}{p^3}} - \frac{y}{2\sqrt{\frac{1}{p^3}}}\right)(1 + e\,\text{Cos}[f])^2;$$

$$\text{deldot}[p_, e_, f_, y_] := \left(\sqrt{\frac{1}{p^3}} - \frac{y}{2\sqrt{\frac{1}{p^3}}}\right)\frac{3}{p};$$

$$\text{pdot}[p_, e_, y_] := -\frac{8\left(1-e^2\right)^{3/2}\left(8+7e^2\right)}{5p^3} + \left(16\left(-1+e^2\right) + \frac{4\left(-168-283e^2+26e^4\right)}{15\left(1-e^2\right)^{3/2}}\right)y;$$

$$\text{edot}[p_, e_, y_] := -\frac{e\left(1-e^2\right)^{3/2}\left(304+121e^2\right)}{15p^4} + \left(\frac{2\left(-24-965e^2-784e^4+73e^6\right)}{15e\left(1-e^2\right)^{3/2}p} + \frac{32\left(1-e^2\right)^{3/2}\left(-1+\sqrt{1-e^2}\right)}{3ep}\right)y;$$

$$\text{xpos} = \frac{p[t]}{1+e[t]\,\text{Cos}[f[t]]}\,\text{Cos}[f[t]+\delta[t]];$$

$$\text{ypos} = \frac{p[t]}{1+e[t]\,\text{Cos}[f[t]]}\,\text{Sin}[f[t]+\delta[t]];$$

Solvers

$\lambda = 0$ vs $\lambda = 10^{-7}$, p=50, e=0.7

In[ ]:=
```
Clear[sol]
```

In[12]:=
```
sol[p0_(*initial semi latus rectum*), e0_(*initial ecentricity*), y_(*sds parameter*)] :=
 Module[{pInit, eInit, yInit},
  pInit = p0;
  eInit = e0;
  yInit = y;
  NDSolve[{p'[t] == pdot[p[t], e[t], y], e'[t] == edot[p[t], e[t], y],
     f'[t] == fdot[p[t], e[t], f[t], y], δ'[t] == deldot[p[t], e[t], f[t], y], f[0] == 0,
     e[0] == eInit, p[0] == pInit, δ[0] == 0, WhenEvent[p[t] == 6 + 2 e[t], "StopIntegration"]
     (*stop at near separatrix*)}, {p, e, f, δ}, {t, 0, 10^6 (*cycles*)}][[1]]]
```

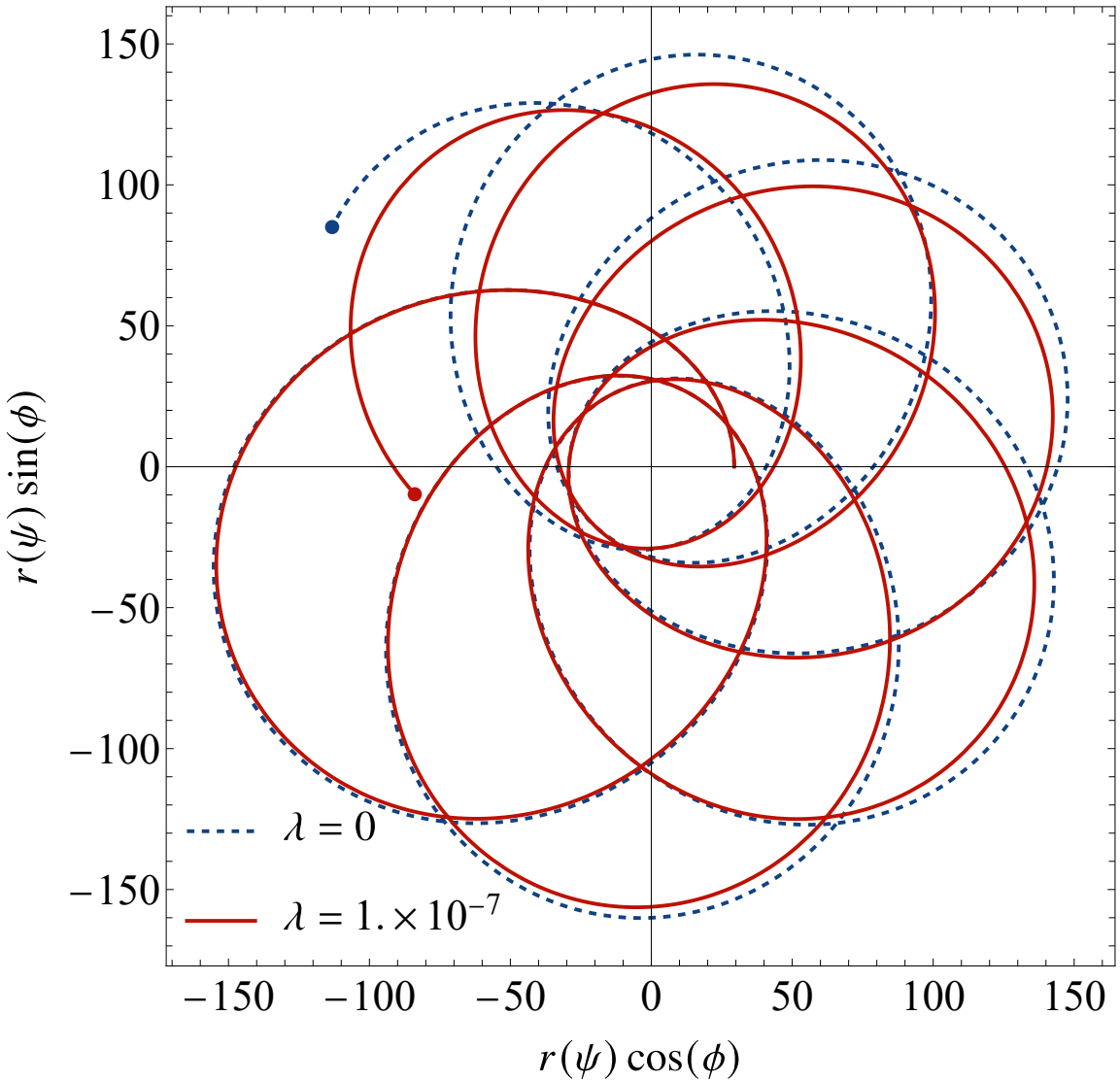
In[ ]:=
```
sol1 = sol[50, 0.7, 0];
sol2 = sol[50, 0.7, 10^-7];
```

Trajectory up to final time tf

In[29]:=
```
wfsdsorb[p0_, e0_, y0_, tf0_] := Module[{tf, p, e, y},
  p = p0;
  e = p0;
  y = y0;
  tf = tf0;
  ParametricPlot[{Evaluate[{xpos, ypos} /. sol[p0, e0, 0]],
    Evaluate[{xpos, ypos} /. sol[p0, e0, y]]}, {t, 0, tf}, Frame → True,
   ImageSize → 600, PlotStyle → {{Dashed, Thick, ■}, {Thick, ■}}, LabelStyle → style,
   FrameLabel → {MaTeX["r(\\psi) \\cos(\\phi)"], MaTeX["r(\\psi) \\sin(\\phi)"]},
   Epilog → {{PointSize → 0.015, ■, Point[{xpos /. sol[p0, e0, 0] /. t → tf,
        ypos /. sol[p0, e0, 0] /. t → tf}]}, {PointSize → 0.015, ■,
      Point[{xpos /. sol[p0, e0, y0] /. t → tf, ypos /. sol[p0, e0, y0] /. t → tf}]}},
   PlotLegends → Placed[{MaTeX["\\lambda=0"], MaTeX["\\lambda="] ×
       TraditionalForm[y0 // N]}, {Left, Bottom}]]]
```

In[30]:=
```
wfsdsorb1 = wfsdsorb[50, 0.7, 10^-7, 31000]
```

Out[30]=

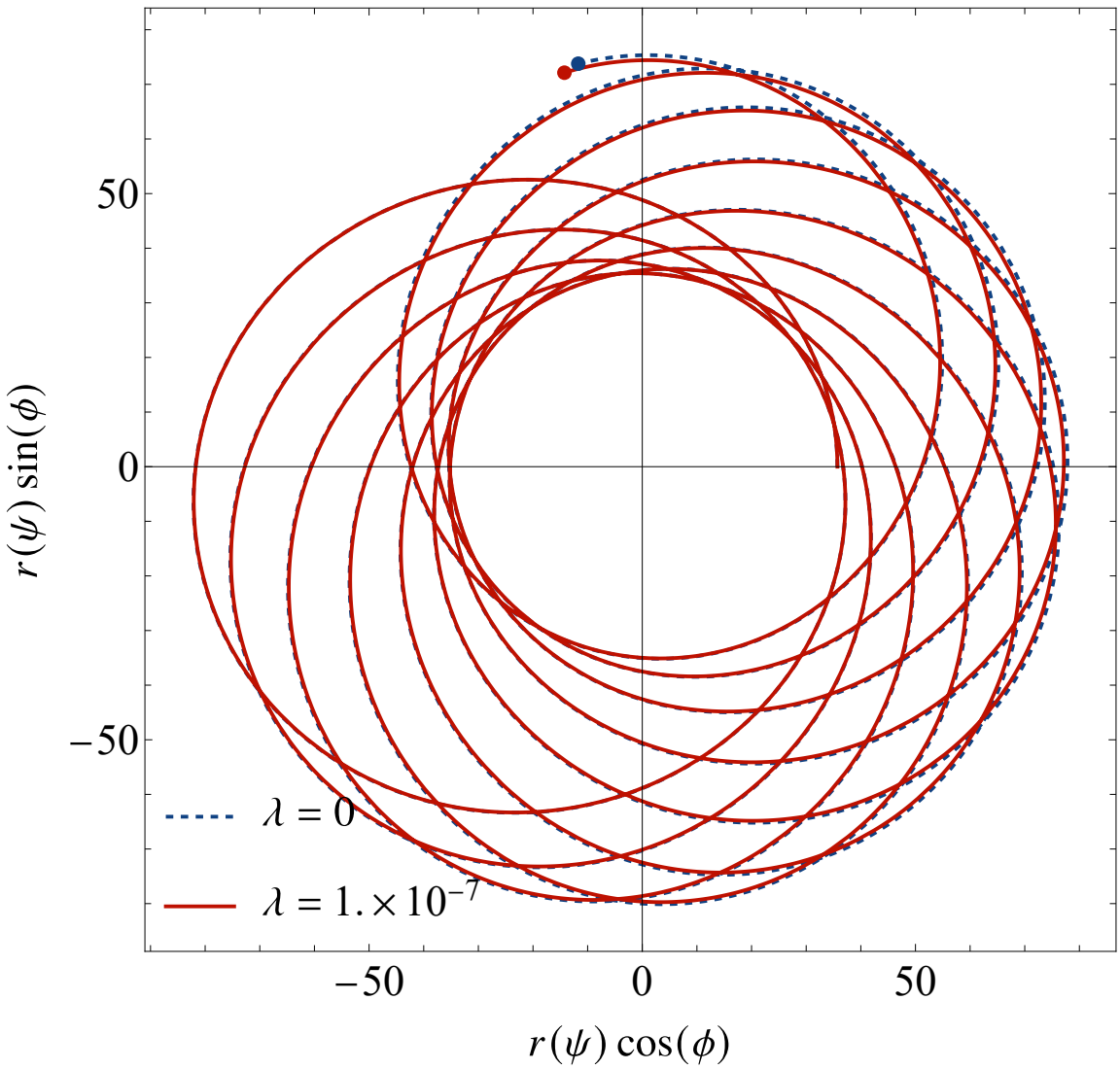


In[31]:=
```
exportPath = FileNameJoin[{plotsFolder, "orbit_50_7_-7.pdf"}];
Export[exportPath, wfsdsorb1];
```

$\lambda = 0$ vs $\lambda = 10^{-7}$, p=50, e=0.4

In[39]:=
```
wfsdsorb2 = wfsdsorb[50, 0.4, 10^-7, 26000]
```

Out[39]=

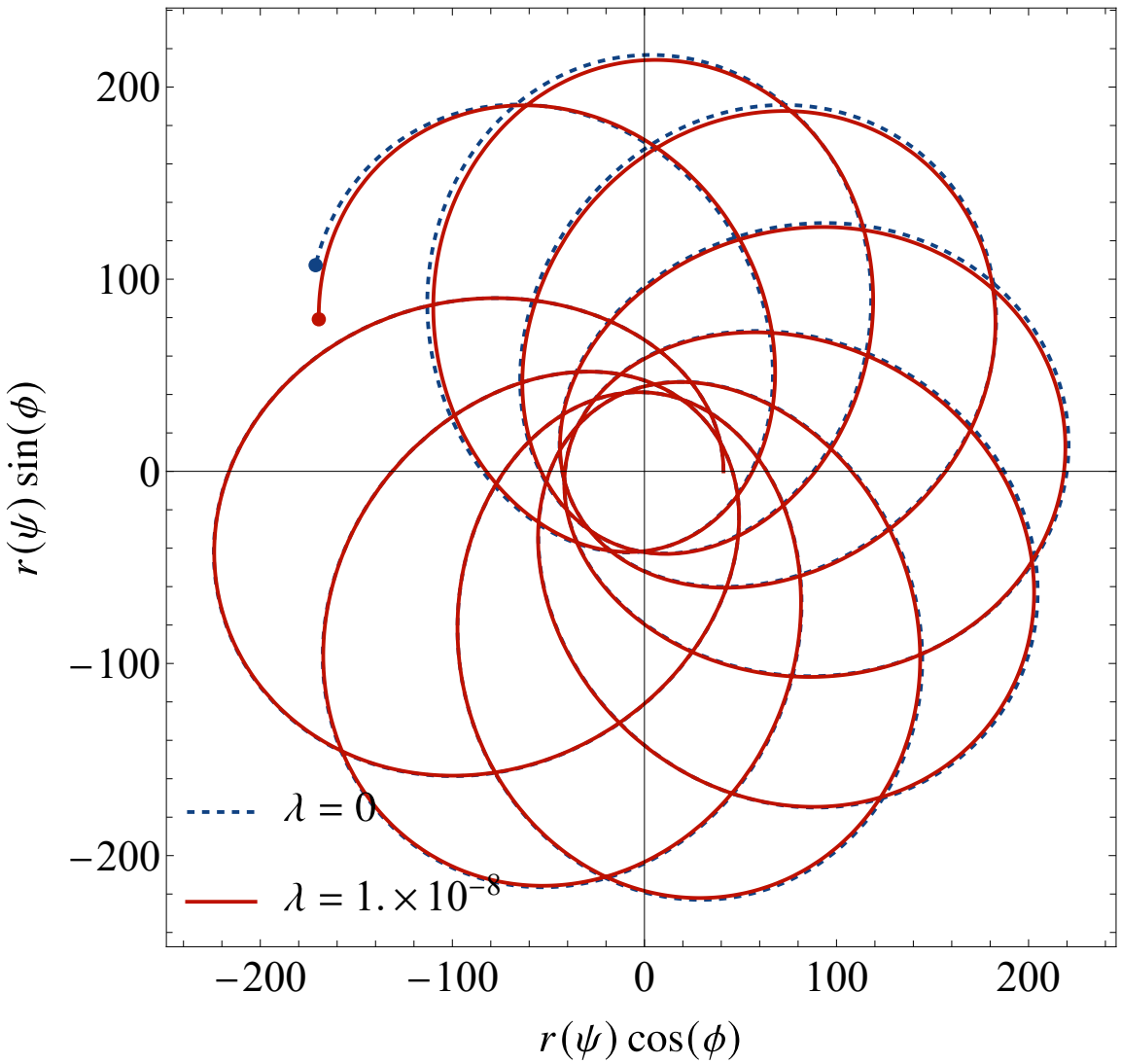


In[36]:=
```
exportPath = FileNameJoin[{plotsFolder, "orbit_50_4_-7.pdf"}];
Export[exportPath, wfsdsorb2];
```

$\lambda = 0$ vs $\lambda = 10^{-8}$, p=70, e=0.7

In[40]:=
```
wfsdsorb3 = wfsdsorb[70, 0.7, 10^-8, 7.3 * 10^4]
```

Out[40]=

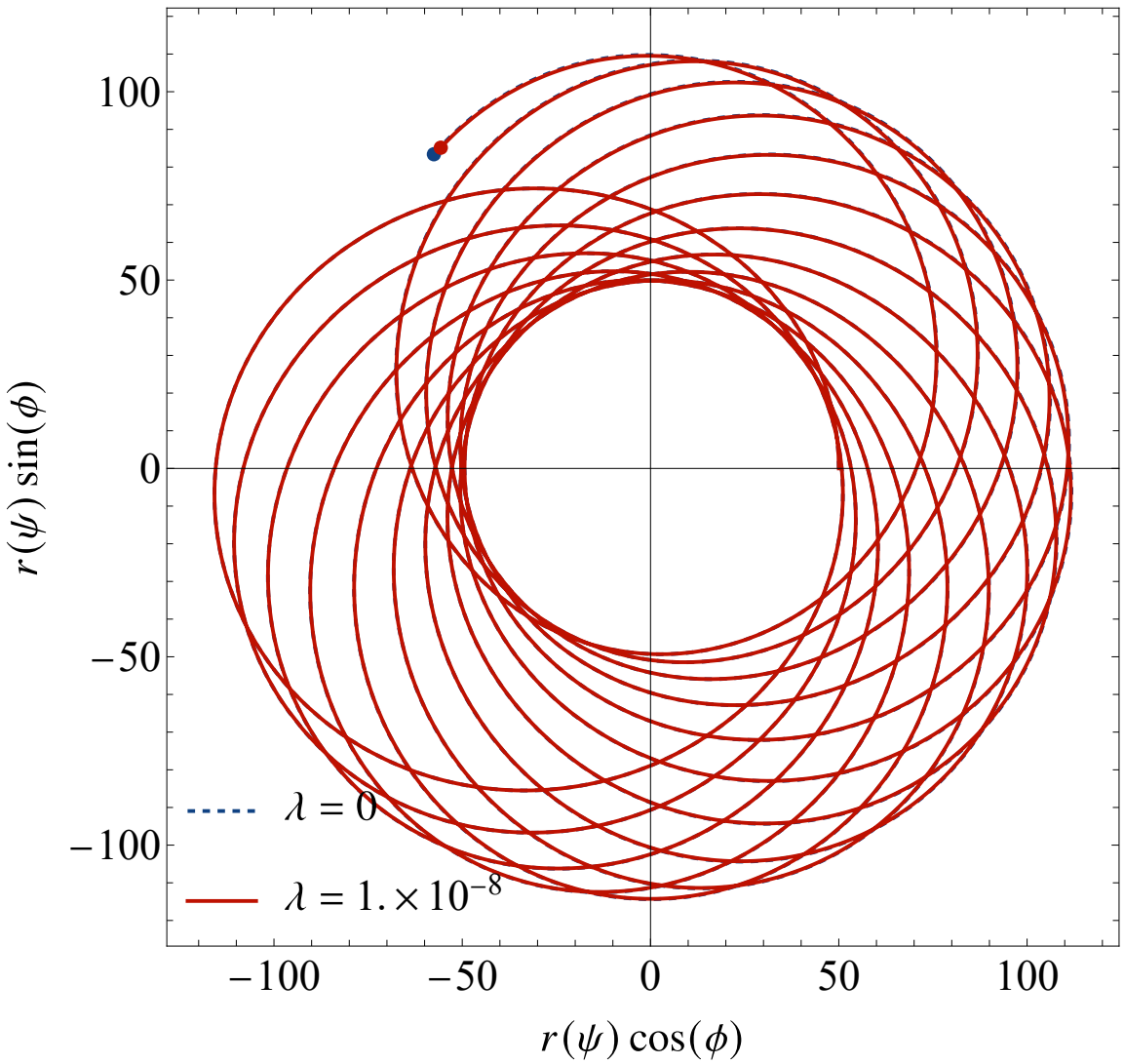


In[41]:=
```
exportPath = FileNameJoin[{plotsFolder, "orbit_70_7_-8.pdf"}];
Export[exportPath, wfsdsorb3];
```

$\lambda = 0$ vs $\lambda = 10^{-8}$, p=70, e=0.4

In[45]:=
```
wfsdsorb4 = wfsdsorb[70, 0.4, 10^-8, 6.3 * 10^4]
```

Out[45]=



In[46]:=
```
exportPath = FileNameJoin[{plotsFolder, "orbit_70_4_-8.pdf"}];
Export[exportPath, wfsdsorb4];
```